

MONTIJO AB, Portugal

WMO: 085340

Lat: 38.7139N

Lon: 9.0245W

Elev: 36

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.6	40.6	31.0	25.3	44.2	34.2	28.9	45.8	24.4	57.6	21.6	55.3	5.9	30	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.7	91.8	69.9	88.7	69.3	85.8	68.7	72.1	86.3	71.0	83.8	69.9	81.6	10.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.2	103.6	74.3	67.0	99.4	74.0	66.1	96.3	73.9	35.7	85.6	34.6	84.1	33.8	82.3	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	18.6	16.5	33.1	100.2	2.9	4.7	31.0	103.6	29.3	106.4	27.7	109.0	25.5	112.4
			32.1	75.3	2.8	1.6	30.0	76.5	28.4	77.4	26.8	78.3	24.8	79.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	52.0	53.7	57.4	60.9	65.6	70.4	73.8	74.6	71.6	66.4	58.0	52.9	(6)
	DBStd	9.03	4.37	3.82	3.82	4.36	4.88	4.47	3.66	3.41	3.39	4.29	4.64	4.56	(7)
	HDD50	65	29	12	1	0	0	0	0	0	0	0	1	22	(8)
	HDD65	1770	402	317	237	139	49	4	0	0	0	34	213	375	(9)
	CDD50	4875	92	115	231	326	485	613	737	763	649	510	242	112	(10)
	CDD65	1103	0	0	1	15	69	168	272	298	199	78	3	0	(11)
	CDH74	7453	0	0	26	120	561	1108	1882	2225	1177	353	1	0	(12)
	CDH80	2486	0	0	1	13	171	385	662	835	342	77	0	0	(13)
(14) Wind	WSAvg	8.5	7.8	8.4	8.8	8.9	9.0	9.1	9.3	8.9	7.8	7.9	8.2	8.1	(14)
(15) Precipitation	PrecAvg	28.9	4.1	3.6	2.8	2.4	1.9	0.7	0.1	0.2	1.2	3.7	4.8	4.0	(15)
	PrecMax	46.8	13.9	11.7	10.5	6.4	6.1	3.9	1.7	2.3	5.1	10.3	14.7	15.1	(16)
	PrecMin	16.4	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	(17)
	PrecStd	8.1	3.0	2.6	2.3	1.6	1.6	0.9	0.3	0.4	1.3	2.9	3.4	3.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	67.8	77.2	82.2	91.2	94.6	97.0	97.1	91.7	87.6	73.2	66.2	(19)
		MCWB	58.4	57.9	60.2	65.8	68.9	70.4	72.1	70.6	68.3	66.2	62.9	61.9	(20)
	2%	DB	62.2	64.6	72.0	77.3	84.3	88.8	91.3	91.4	87.6	81.3	69.9	64.2	(21)
		MCWB	57.9	56.6	59.0	63.3	66.6	69.9	70.5	69.9	68.2	66.2	62.7	60.3	(22)
	5%	DB	60.7	62.6	68.2	73.5	80.2	84.2	86.3	87.8	84.0	77.3	67.6	62.4	(23)
		MCWB	57.0	56.6	58.5	61.9	65.4	68.2	69.3	69.7	67.9	65.3	61.2	58.8	(24)
	10%	DB	59.1	60.8	64.8	69.8	75.6	80.2	82.8	84.5	80.6	74.8	65.4	60.6	(25)
		MCWB	55.7	55.8	57.0	60.5	63.8	67.0	68.4	68.5	67.2	64.9	60.3	57.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	60.7	64.2	67.3	70.9	73.3	73.8	73.4	72.0	70.8	67.1	63.1	(27)
		MCDB	62.3	63.6	72.7	79.3	82.9	90.1	93.1	88.3	83.4	78.4	69.9	65.2	(28)
	2%	WB	59.3	59.3	61.2	64.6	68.4	71.0	71.8	71.9	70.4	69.2	64.7	60.9	(29)
		MCDB	61.1	61.9	68.0	74.7	80.3	84.9	86.9	85.9	81.1	75.1	67.8	63.3	(30)
	5%	WB	58.1	58.1	59.7	62.8	66.5	69.3	70.3	70.7	69.3	68.0	62.9	59.4	(31)
		MCDB	59.9	61.1	65.4	71.4	78.0	81.5	83.5	83.9	79.5	73.4	65.9	61.8	(32)
	10%	WB	56.7	56.6	58.3	60.9	64.5	67.7	69.1	69.5	68.3	66.6	61.3	57.8	(33)
		MCDB	58.7	60.2	63.5	68.4	74.3	78.7	81.2	81.9	77.7	72.4	64.4	60.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	14.2	15.5	15.9	17.7	18.1	19.3	20.7	19.1	16.8	13.6	13.1	(35)
		MCDBR	13.3	16.6	22.1	22.9	25.6	26.3	25.9	26.5	24.9	22.3	15.2	13.1	(36)
	5% WB	MCWBR	9.3	10.4	11.6	11.6	11.6	11.1	10.2	10.3	9.9	10.3	9.1	9.4	(37)
		MCWBR	10.3	13.5	17.6	20.4	22.5	23.1	22.3	22.4	20.1	16.3	11.6	11.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.330	0.344	0.374	0.390	0.397	0.402	0.386	0.399	0.398	0.366	0.341	0.331	(40)
		taud	2.506	2.463	2.388	2.340	2.348	2.356	2.391	2.363	2.379	2.490	2.516	2.505	(41)
	Ebn at Noon		268	279	280	282	280	278	282	275	269	268	262	259	(42)
		Edh at Noon	25	30	35	39	39	39	37	37	35	28	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		715	1030	1397	1792	2152	2339	2390	2164	1690	1160	795	645	(44)
	RadStd		90	121	149	150	149	124	92	81	108	98	99	73	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page